

Solvability conditions for asymmetric indefinite Riccati equation in mean-field game feedback

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Website #38 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Let

$$\hat{A}_t := A_t + \bar{A}_t, \quad \hat{D}_t := D_t + \bar{D}_t, \quad Q_t^\sharp := (1 - \alpha_1)Q_t, \quad \Sigma_T^\sharp := (1 - \alpha_3)L_T.$$

We seek verifiable *sufficient* conditions on the deterministic coefficient data

$$A, \bar{A}, D, \bar{D}, B, \bar{B}, F, \bar{F}, Q, R, S, L_T$$

(and any scalar parameters such as $\alpha_1, \alpha_3, \beta_1$ entering the auxiliary matrices) ensuring that the terminal-value matrix ODE

$$\dot{\Sigma}_t + \Sigma_t \hat{A}_t + A_t^\top \Sigma_t + D_t^\top \Sigma_t \hat{D}_t - \Sigma_{e,t} R_{e,t}^{-1} \bar{\Sigma}_t^\top + Q_t^\sharp = 0, \quad \Sigma_T = \Sigma_T^\sharp,$$

admits a unique sufficiently regular solution $\Sigma : [0, T] \rightarrow \mathbb{R}^{n \times n}$, where $R_{e,t}, \Sigma_{e,t}, \bar{\Sigma}_t$ are affine in Σ_t , and such that $R_{e,t}$ stays uniformly invertible on $[0, T]$.

Equivalently: write the auxiliary matrices as affine maps

$$R_e(t, X), \quad \Sigma_e(t, X), \quad \bar{\Sigma}(t, X),$$

with $X \in \mathbb{R}^{n \times n}$, and find conditions guaranteeing a unique Σ solving

$$\dot{\Sigma}_t + F(t, \Sigma_t) = 0, \quad \Sigma_T = \Sigma_T^\sharp,$$

where

$$F(t, X) := X \hat{A}_t + A_t^\top X + D_t^\top X \hat{D}_t - \Sigma_e(t, X) R_e(t, X)^{-1} \bar{\Sigma}(t, X)^\top + Q_t^\sharp.$$

The negation is: there exist admissible coefficients for which either no absolutely continuous solution exists on all of $[0, T]$, or there are two distinct such solutions, or every candidate solution forces $R_e(t, \Sigma_t)$ to become singular somewhere on $[0, T]$.

Alternative readings considered

The statement leaves the exact formulas of $R_e, \Sigma_e, \bar{\Sigma}$ implicit. I considered two readings:

- *Hard reading*: give conditions for an arbitrary affine dependence on Σ .
- *Concrete-paper reading*: plug in one specific paper's formulas.

I adopt the hard reading. Any concrete model from the paper is obtained by expanding its affine formulas into the abstract affine form used below. If the concrete formulas also contain an already-solved symmetric Riccati matrix Π , I treat Π as part of the frozen deterministic coefficient data, since the present question isolates the Σ -equation.

Related literature found

- **Li–Nie–Wang–Yan (2024)**, *Journal of Optimization Theory and Applications* — PARTIAL. This is the closest source I found: incomplete-information LQ mean-field games with common noise and control-dependent diffusions, two Riccati equations, and a solvability result for the asymmetric one only under extra structural assumptions that reduce it to a standard Riccati problem. (arxiv.org)
- **Li–Nie–Wu (2023)**, *SIAM Journal on Control and Optimization* — PARTIAL. Studies a partial-information large-population LQ problem with control entering the diffusion and states that in the unconstrained case the decentralized strategy is represented by a Riccati approach, with existence/uniqueness for a new Riccati-type consistency system discussed. (epubs.siam.org)
- **Bensoussan–Feng–Huang (2021)**, *Mathematical Control and Related Fields* — BACKGROUND. Earlier LQG mean-field game with partial observation and common noise; the consistency condition is connected to a common-noise FBSDE, but the diffusion structure is simpler than in the present asymmetric rational Riccati equation. (ira.lib.polyu.edu.hk)
- **Ait Rami–Moore–Zhou (2002)**, *SIAM Journal on Control and Optimization* — TOOL. Shows that indefinite stochastic LQ problems may require generalized Riccati equations with algebraic constraints; methodologically this warns that one cannot hope for a completely sign-free global theorem without extra effective convexity/invertibility conditions. (epubs.siam.org)
- **Freiling (2002)**, *Linear Algebra and its Applications* — BACKGROUND. Survey of non-symmetric Riccati equations and their applications; useful for orienting the asymmetric part of the problem. (sciencedirect.com)
- **Freiling–Jank–Sarychev (2000)**, *Results in Mathematics* — TOOL. Develops non-blow-up / invariant-set ideas for nonsymmetric Riccati equations; I do not import a theorem from it, but its viewpoint strongly suggests the invariant-tube argument used below. (link.springer.com)
- **Liu–Firoozi (2026)**, *arXiv* — BACKGROUND. Promising because it treats common-noise LQ mean-field games on small and arbitrary horizons, but its object is a coupled linear FBSEE in Hilbert space, not the finite-dimensional rational asymmetric Riccati ODE here. (arxiv.org)

Proof sketch

Because the auxiliary matrices are only specified to be affine in Σ , the right framework is to regard the Riccati equation as an ODE on the Banach space $\mathbb{R}^{n \times n}$ with a rational nonlinearity

$$X \mapsto \Sigma_e(t, X) R_e(t, X)^{-1} \bar{\Sigma}(t, X)^\top.$$

The two genuine obstructions are:

1. $R_e(t, X)$ may lose invertibility.
2. Even if invertible, the rational term may grow too fast to keep the solution in a controlled region.

So the right object is an *invariant tube*

$$\{(t, X) : \|X\| \leq M, \text{sym } R_e(t, X) \geq rI_m\}.$$

Inside such a tube, the inverse map is uniformly bounded and Lipschitz, hence the whole vector field is bounded and Lipschitz. This yields a generic full-interval theorem by a single backward contraction on $C([0, T]; \mathbb{R}^{n \times n})$. The price is a small-gain condition involving T , the terminal norm, and the affine slopes.

That criterion is robust and fully asymmetric, but it is only sufficient. To go beyond small horizons, I prove a second criterion: if the rational term enjoys a *linear-growth estimate* on a larger positivity tube, then one can solve the equation on short subintervals by contraction and concatenate the solutions, while a scalar comparison argument keeps the norm strictly below the invertibility threshold. This yields an arbitrary-horizon theorem under a cancellation/dissipativity hypothesis on the concrete affine formulas.

The proof is self-contained: the only analytic input is a contraction argument, reproved below.

Main result

Let $\mathbb{X} := \mathbb{R}^{n \times n}$, equipped with the spectral norm $\|\cdot\|$. For a matrix M , write

$$\text{sym } M := \frac{1}{2}(M + M^\top).$$

For a linear map $\mathcal{T} : \mathbb{X} \rightarrow \mathbb{R}^{p \times q}$, write

$$\|\mathcal{T}\| := \sup_{\|X\|=1} \|\mathcal{T}[X]\|.$$

Assume that the affine auxiliary matrices admit the decomposition

$$R_e(t, X) = R_{0,t} + \mathcal{R}_t[X], \quad \Sigma_e(t, X) = U_{0,t} + \mathcal{U}_t[X], \quad \bar{\Sigma}(t, X) = V_{0,t} + \mathcal{V}_t[X],$$

where $R_{0,\cdot}, U_{0,\cdot}, V_{0,\cdot}$ are bounded measurable matrix functions and $\mathcal{R}_t, \mathcal{U}_t, \mathcal{V}_t$ are bounded measurable linear maps. Any scalar model parameters (including β_1) are absorbed into these coefficients.

Define

$$a := \|\hat{A}\|_{L^\infty(0,T)} + \|A\|_{L^\infty(0,T)} + \|D\|_{L^\infty(0,T)} \|\hat{D}\|_{L^\infty(0,T)},$$

$$q := \|Q^\sharp\|_{L^\infty(0,T)}, \quad \ell := \|\Sigma_T^\sharp\|,$$

$$r_0 := \text{ess inf}_{t \in [0,T]} \lambda_{\min}(\text{sym } R_{0,t}),$$

$$\rho_R := \|\mathcal{R}\|_{L^\infty(0,T)}, \quad u_0 := \|U_0\|_{L^\infty(0,T)}, \quad v_0 := \|V_0\|_{L^\infty(0,T)},$$

$$\rho_U := \|\mathcal{U}\|_{L^\infty(0,T)}, \quad \rho_V := \|\mathcal{V}\|_{L^\infty(0,T)}.$$

For $M > 0$, set

$$r_M := r_0 - \rho_R M, \quad u_M := u_0 + \rho_U M, \quad v_M := v_0 + \rho_V M,$$

$$N_M := \frac{u_M v_M}{r_M},$$

$$\Lambda_M := \frac{\rho_U v_M + u_M \rho_V}{r_M} + \frac{u_M v_M \rho_R}{r_M^2}, \quad L_M := a + \Lambda_M.$$

Theorem 1 (Two explicit sufficient criteria). *Assume the affine decomposition above.*

(i) **Generic full-interval criterion (single contraction).** *If there exists $M > \ell$ such that*

$$r_M > 0,$$

$$\ell + T(aM + q + N_M) \leq M,$$

$$T L_M < 1,$$

then the asymmetric Riccati equation admits a unique solution

$$\Sigma \in C([0, T]; \mathbb{X}) \cap W^{1, \infty}(0, T; \mathbb{X})$$

with

$$\sup_{t \in [0, T]} \|\Sigma_t\| \leq M, \quad \text{sym } R_e(t, \Sigma_t) \geq r_M I_m \quad \text{for a.e. } t \in [0, T].$$

Hence $R_e(t, \Sigma_t)$ is uniformly invertible and

$$\|R_e(t, \Sigma_t)^{-1}\| \leq r_M^{-1} \quad \text{for a.e. } t.$$

(ii) **Arbitrary-horizon criterion by continuation.** *Suppose there exist $M > 0$ and constants $\gamma_0, \gamma_1 \geq 0$ such that*

$$r_M > 0,$$

and for all $t \in [0, T]$ and all $X \in \mathbb{X}$ with $\|X\| \leq M$,

$$\|\Sigma_e(t, X) R_e(t, X)^{-1} \bar{\Sigma}(t, X)^\top\| \leq \gamma_0 + \gamma_1 \|X\|.$$

Set

$$c_0 := q + \gamma_0, \quad c_1 := a + \gamma_1,$$

and define the scalar comparison function

$$g(t) := \begin{cases} e^{c_1(T-t)} \ell + \frac{c_0}{c_1} (e^{c_1(T-t)} - 1), & c_1 > 0, \\ \ell + c_0(T - t), & c_1 = 0. \end{cases}$$

If

$$g(0) < M,$$

then the Riccati equation admits a unique solution

$$\Sigma \in C([0, T]; \mathbb{X}) \cap W^{1, \infty}(0, T; \mathbb{X})$$

satisfying

$$\|\Sigma_t\| \leq g(t) \quad \forall t \in [0, T],$$

and therefore

$$\text{sym } R_e(t, \Sigma_t) \geq (r_0 - \rho_R g(0)) I_m \quad \text{for a.e. } t \in [0, T].$$

In particular $R_e(t, \Sigma_t)$ is uniformly invertible on $[0, T]$.

In either case, if the coefficient functions are continuous in t , then the solution is in fact C^1 . Consequently any decentralized linear feedback gain built affinely from the fixed data, Σ , and already-solved bounded coefficients is bounded on $[0, T]$; thus the feedback representation is well-posed.

Proof

Lemma 1 (Coefficient bookkeeping). *Suppose an affine map $\mathcal{T}_t : \mathbb{X} \rightarrow \mathbb{R}^{p \times q}$ has the concrete form*

$$\mathcal{T}_t[X] = T_{0,t} + \sum_{j=1}^J M_{j,t} X N_{j,t}.$$

Then

$$\|\mathcal{T}_t[X]\| \leq \|T_{0,t}\| + \left(\sum_{j=1}^J \|M_{j,t}\| \|N_{j,t}\| \right) \|X\|.$$

Hence

$$\|\mathcal{T}\|_{L^\infty(0,T)} \leq \left\| \sum_{j=1}^J \|M_{j,\cdot}\| \|N_{j,\cdot}\| \right\|_{L^\infty(0,T)}.$$

Proof. By submultiplicativity,

$$\|M_{j,t} X N_{j,t}\| \leq \|M_{j,t}\| \|X\| \|N_{j,t}\|.$$

Summing over j gives the claim. □

Lemma 2 (Invertibility on a positivity tube). *Fix $M > 0$ with $r_M > 0$. Then for every $t \in [0, T]$ and every $X, Y \in \mathbb{X}$ with $\|X\|, \|Y\| \leq M$,*

$$\text{sym } R_e(t, X) \geq r_M I_m,$$

hence $R_e(t, X)$ is invertible and

$$\|R_e(t, X)^{-1}\| \leq r_M^{-1}.$$

Moreover,

$$\|R_e(t, X)^{-1} - R_e(t, Y)^{-1}\| \leq \frac{\rho_R}{r_M^2} \|X - Y\|.$$

Proof. Since $\|\mathcal{R}_t[X]\| \leq \rho_R \|X\| \leq \rho_R M$,

$$\lambda_{\min}(\text{sym } R_e(t, X)) \geq \lambda_{\min}(\text{sym } R_{0,t}) - \|\text{sym } \mathcal{R}_t[X]\| \geq r_0 - \rho_R M = r_M.$$

Now let $z \in \mathbb{R}^m$. Then

$$r_M \|z\|^2 \leq z^\top \text{sym } R_e(t, X) z = z^\top R_e(t, X) z \leq \|R_e(t, X) z\| \|z\|,$$

so $\|R_e(t, X) z\| \geq r_M \|z\|$ for every z . Therefore $R_e(t, X)$ is invertible and $\|R_e(t, X)^{-1}\| \leq r_M^{-1}$.

For the difference of inverses,

$$R_e(t, X)^{-1} - R_e(t, Y)^{-1} = R_e(t, X)^{-1} (R_e(t, Y) - R_e(t, X)) R_e(t, Y)^{-1},$$

hence

$$\|R_e(t, X)^{-1} - R_e(t, Y)^{-1}\| \leq r_M^{-2} \|\mathcal{R}_t[Y - X]\| \leq \frac{\rho_R}{r_M^2} \|X - Y\|.$$

□

Lemma 3 (Bounds and Lipschitz constant for the vector field). *Fix $M > 0$ with $r_M > 0$. Let*

$$N(t, X) := \Sigma_e(t, X) R_e(t, X)^{-1} \bar{\Sigma}(t, X)^\top.$$

Then for all $t \in [0, T]$ and $\|X\|, \|Y\| \leq M$,

$$\|\Sigma_e(t, X)\| \leq u_M, \quad \|\bar{\Sigma}(t, X)\| \leq v_M,$$

$$\|N(t, X)\| \leq N_M,$$

and

$$\|N(t, X) - N(t, Y)\| \leq \Lambda_M \|X - Y\|.$$

Consequently, with

$$F(t, X) := X \hat{A}_t + A_t^\top X + D_t^\top X \hat{D}_t - N(t, X) + Q_t^\sharp,$$

one has

$$\|F(t, X)\| \leq aM + q + N_M,$$

$$\|F(t, X) - F(t, Y)\| \leq L_M \|X - Y\|.$$

Proof. The first bound is immediate from affinity:

$$\|\Sigma_e(t, X)\| \leq \|U_{0,t}\| + \|\mathcal{U}_t\| \|X\| \leq u_0 + \rho_U M = u_M,$$

and similarly $\|\bar{\Sigma}(t, X)\| \leq v_M$.

By the previous lemma,

$$\|N(t, X)\| \leq \|\Sigma_e(t, X)\| \|R_e(t, X)^{-1}\| \|\bar{\Sigma}(t, X)\| \leq \frac{u_M v_M}{r_M} = N_M.$$

For the difference, write

$$N(t, X) - N(t, Y) = I_1 + I_2 + I_3,$$

where

$$\begin{aligned} I_1 &:= (\Sigma_e(t, X) - \Sigma_e(t, Y)) R_e(t, X)^{-1} \bar{\Sigma}(t, X)^\top, \\ I_2 &:= \Sigma_e(t, Y) (R_e(t, X)^{-1} - R_e(t, Y)^{-1}) \bar{\Sigma}(t, X)^\top, \\ I_3 &:= \Sigma_e(t, Y) R_e(t, Y)^{-1} (\bar{\Sigma}(t, X) - \bar{\Sigma}(t, Y))^\top. \end{aligned}$$

Then

$$\|I_1\| \leq \rho_U \|X - Y\| r_M^{-1} v_M,$$

$$\|I_2\| \leq u_M \frac{\rho_R}{r_M^2} \|X - Y\| v_M,$$

$$\|I_3\| \leq u_M r_M^{-1} \rho_V \|X - Y\|.$$

Summing gives the Lipschitz constant Λ_M .

Finally,

$$\|X \hat{A}_t + A_t^\top X + D_t^\top X \hat{D}_t\| \leq (\|\hat{A}_t\| + \|A_t\| + \|D_t\| \|\hat{D}_t\|) \|X\| \leq aM,$$

hence the bounds for F . □

Lemma 4 (Contraction principle). *Let $(\mathcal{X}, d)$ be a complete metric space and let $\Gamma : \mathcal{X} \rightarrow \mathcal{X}$ satisfy*

$$d(\Gamma x, \Gamma y) \leq \kappa d(x, y) \quad \forall x, y \in \mathcal{X}$$

for some $\kappa \in [0, 1)$. Then Γ has a unique fixed point.

Proof. Pick $x_0 \in \mathcal{X}$ and define $x_{k+1} := \Gamma x_k$. Then

$$d(x_{k+1}, x_k) \leq \kappa^k d(x_1, x_0).$$

Hence for $m > n$,

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \leq d(x_1, x_0) \sum_{j=n}^{m-1} \kappa^j.$$

The geometric tail tends to 0, so (x_k) is Cauchy. Completeness gives $x_k \rightarrow x_* \in \mathcal{X}$. Since Γ is Lipschitz, hence continuous,

$$\Gamma x_* = \Gamma \left(\lim_{k \rightarrow \infty} x_k \right) = \lim_{k \rightarrow \infty} \Gamma x_k = \lim_{k \rightarrow \infty} x_{k+1} = x_*.$$

If y_* is another fixed point, then

$$d(x_*, y_*) = d(\Gamma x_*, \Gamma y_*) \leq \kappa d(x_*, y_*),$$

so $(1 - \kappa)d(x_*, y_*) \leq 0$, hence $x_* = y_*$. □

Proof of the theorem. Part (i). Let

$$\mathcal{B}_M := \left\{ X \in C([0, T]; \mathbb{X}) : \|X\|_\infty := \sup_{t \in [0, T]} \|X_t\| \leq M \right\}.$$

This is a complete metric space for the sup norm.

Define the backward integral operator

$$(\Gamma X)(t) := \Sigma_T^\# + \int_t^T F(s, X_s) ds.$$

For $X \in \mathcal{B}_M$, Lemma 3 gives

$$\|(\Gamma X)(t)\| \leq \ell + \int_t^T (aM + q + N_M) ds \leq \ell + T(aM + q + N_M) \leq M.$$

Hence $\Gamma(\mathcal{B}_M) \subset \mathcal{B}_M$.

If $X, Y \in \mathcal{B}_M$, then by Lemma 3,

$$\|(\Gamma X) - (\Gamma Y)\|_\infty \leq \sup_{t \in [0, T]} \int_t^T L_M \|X_s - Y_s\| ds \leq TL_M \|X - Y\|_\infty.$$

Since $TL_M < 1$, Lemma 4 yields a unique fixed point $\Sigma \in \mathcal{B}_M$.

By construction,

$$\Sigma_t = \Sigma_T^\# + \int_t^T F(s, \Sigma_s) ds,$$

so Σ is absolutely continuous, $\Sigma_T = \Sigma_T^\sharp$, and

$$\dot{\Sigma}_t + F(t, \Sigma_t) = 0 \quad \text{for a.e. } t \in [0, T].$$

Moreover $\|F(t, \Sigma_t)\| \leq aM + q + N_M$, so $\Sigma \in W^{1,\infty}(0, T; \mathbb{X})$. Lemma 2 gives

$$\text{sym } R_e(t, \Sigma_t) \geq r_M I_m,$$

hence uniform invertibility. If the coefficients are continuous in t , then $t \mapsto F(t, \Sigma_t)$ is continuous and $\Sigma \in C^1$.

Part (ii). Choose M_1 such that

$$g(0) < M_1 < M.$$

Then $r_{M_1} > 0$. Let L_{M_1} be the constant from Lemma 3 corresponding to M_1 . Choose

$$0 < \delta < \min\left\{\frac{1}{L_{M_1}}, \frac{M_1 - g(0)}{c_0 + c_1 M_1}\right\}.$$

Take a partition

$$0 = t_0 < t_1 < \dots < t_N = T$$

with $t_k - t_{k-1} \leq \delta$ for every k .

We construct the solution backward by induction on $k = N, N-1, \dots, 1$.

Assume that a terminal matrix Y_k at time t_k is given with

$$\|Y_k\| \leq g(t_k).$$

Consider

$$\mathcal{B}_{M_1}^{(k)} := \{X \in C([t_{k-1}, t_k]; \mathbb{X}) : \|X\|_\infty \leq M_1\},$$

and define

$$(\Gamma_k X)(t) := Y_k + \int_t^{t_k} F(s, X_s) ds.$$

For $X \in \mathcal{B}_{M_1}^{(k)}$, the assumed linear-growth bound on the rational term implies

$$\|F(s, X_s)\| \leq a\|X_s\| + q + \gamma_0 + \gamma_1\|X_s\| \leq c_0 + c_1 M_1.$$

Hence

$$\|(\Gamma_k X)(t)\| \leq \|Y_k\| + \delta(c_0 + c_1 M_1) \leq g(t_k) + \delta(c_0 + c_1 M_1) \leq g(0) + \delta(c_0 + c_1 M_1) < M_1.$$

So Γ_k maps $\mathcal{B}_{M_1}^{(k)}$ to itself.

Also, by Lemma 3,

$$\|\Gamma_k X - \Gamma_k Z\|_\infty \leq (t_k - t_{k-1})L_{M_1}\|X - Z\|_\infty \leq \delta L_{M_1}\|X - Z\|_\infty,$$

and $\delta L_{M_1} < 1$. Therefore Γ_k has a unique fixed point $X^{(k)}$ on $[t_{k-1}, t_k]$.

It remains to prove the a priori bound

$$\|X_t^{(k)}\| \leq g(t) \quad \forall t \in [t_{k-1}, t_k].$$

For $t \in [t_{k-1}, t_k]$,

$$\|X_t^{(k)}\| \leq \|Y_k\| + \int_t^{t_k} (c_0 + c_1\|X_s^{(k)}\|) ds.$$

Define

$$w(t) := \|Y_k\| + \int_t^{t_k} (c_0 + c_1 \|X_s^{(k)}\|) ds.$$

Then $\|X_t^{(k)}\| \leq w(t)$, $w(t_k) = \|Y_k\|$, and

$$-w'(t) = c_0 + c_1 \|X_t^{(k)}\| \leq c_0 + c_1 w(t) \quad \text{for a.e. } t.$$

If $c_1 > 0$, then

$$-\frac{d}{dt} \left(e^{-c_1 t} w(t) \right) \leq c_0 e^{-c_1 t},$$

and integrating from t to t_k gives

$$w(t) \leq e^{c_1(t_k-t)} \|Y_k\| + \frac{c_0}{c_1} (e^{c_1(t_k-t)} - 1).$$

If $c_1 = 0$, the same calculation gives $w(t) \leq \|Y_k\| + c_0(t_k - t)$.

Since $\|Y_k\| \leq g(t_k)$, the right-hand side is bounded by the unique scalar solution of

$$-\dot{z} = c_0 + c_1 z, \quad z(t_k) = g(t_k),$$

namely $z(t) = g(t)$ on $[t_{k-1}, t_k]$. Therefore

$$\|X_t^{(k)}\| \leq g(t) \quad \forall t \in [t_{k-1}, t_k].$$

In particular,

$$Y_{k-1} := X_{t_{k-1}}^{(k)}$$

satisfies $\|Y_{k-1}\| \leq g(t_{k-1})$, so the induction continues.

Starting from $Y_N = \Sigma_T^\sharp$, with $\|Y_N\| = \ell = g(T)$, the induction constructs a solution on every subinterval, hence on all of $[0, T]$. The pieces glue continuously and are absolutely continuous; since there are only finitely many pieces, the global solution belongs to $W^{1,\infty}(0, T; \mathbb{X})$.

Now let $\tilde{\Sigma}$ be any other solution with $\|\tilde{\Sigma}\|_\infty \leq M$. Repeating the same scalar comparison on each subinterval yields

$$\|\tilde{\Sigma}_t\| \leq g(t) \leq g(0) < M_1,$$

so $\tilde{\Sigma}$ also lies in the contraction class $\mathcal{B}_{M_1}^{(k)}$ on every subinterval. By uniqueness of the fixed point on $[t_{N-1}, T]$, the two solutions coincide there; then the same argument propagates backward interval by interval. Thus the solution is unique in the class $\|X\|_\infty \leq M$.

Finally, since $\|\Sigma_t\| \leq g(0)$,

$$\text{sym } R_e(t, \Sigma_t) \geq (r_0 - \rho_R g(0)) I_m,$$

so $R_e(t, \Sigma_t)$ is uniformly invertible. Continuity in t again upgrades $W^{1,\infty}$ to C^1 .

The last sentence of the theorem follows because the feedback gains in the paper are rational/affine expressions in the deterministic coefficients and $R_e(t, \Sigma_t)^{-1}$; boundedness of Σ and of R_e^{-1} therefore yields bounded deterministic gain matrices. $\square$

Gap to the full statement

The theorem gives *sufficient* conditions, not a necessary-and-sufficient characterization.

More precisely, what remains open in this manuscript is the following sharper problem: characterize solvability on arbitrary finite horizons for the fully general asymmetric indefinite rational Riccati equation *without* either

- the small-gain / invariant-ball conditions of part (i), or
- the linear-growth cancellation condition of part (ii).

In particular, my result does not cover all coefficient sets with indefinite R_t for which the effective matrix $R_e(t, \Sigma_t)$ may still remain invertible by more delicate generalized-Riccati/LMI mechanisms. It also does not exploit special algebraic identities from one specific paper beyond the abstract fact that the auxiliary matrices depend affinely on Σ .

Self red-team

Counterexample attempts.

I searched first in the scalar constant-coefficient case $n = m = 1$, because any flaw should appear there first.

- *Lemma 2 (positivity tube) without strict margin.* Take $R_e(\sigma) = \sigma$. At terminal data $\sigma_T = 0$, the equation is not even defined. So the strict positivity requirement $r_M > 0$ cannot be dropped.
- *Lemma 3 (Lipschitz inverse) without positivity tube.* Again with $R_e(\sigma) = \sigma$, the inverse $1/\sigma$ is not Lipschitz near 0. So the estimate

$$\|R_e^{-1}(X) - R_e^{-1}(Y)\| \leq C\|X - Y\|$$

really needs a uniform invertibility margin.

- *Main theorem: are the conditions necessary?* No. Take

$$R_e(\sigma) = 1, \quad \Sigma_e(\sigma) = \bar{\Sigma}(\sigma) = \sigma, \quad A = \bar{A} = D = \bar{D} = Q = 0.$$

Then

$$\dot{\sigma} - \sigma^2 = 0, \quad \sigma_T = \ell,$$

whose exact solution is

$$\sigma_t = \frac{\ell}{1 + \ell(T - t)}.$$

This exists uniquely for every finite T , even when the small-gain condition $TL_M < 1$ fails. So my criteria are genuinely only sufficient.

Three likely referee objections and responses.

1. *Objection: R_e is not symmetric, so why does positivity of its symmetric part imply invertibility?*
Response: this is proved explicitly in Lemma 2 via

$$z^\top R_e z = z^\top \text{sym}(R_e) z$$

and Cauchy–Schwarz.

2. *Objection: uniqueness is only shown in a bounded class.* Response: yes, intentionally. The theorem states uniqueness in the natural invariant class where the feedback law is meaningful. In part (ii), every bounded solution in the larger tube automatically contracts into the smaller class by the scalar comparison estimate.
3. *Objection: the theorem is “abstract in $R_e, \Sigma_e, \bar{\Sigma}$ ” rather than directly in $B, \bar{B}, F, \bar{F}, S, \beta_1$.* Response: because the exact formulas were not included in the prompt. Lemma 1 gives the missing translation step: once the concrete affine formulas are expanded into sums $M_t X N_t$, all constants $\rho_R, \rho_U, \rho_V, u_0, v_0$ become explicit L^∞ -norm combinations of the given coefficients and scalar parameters.

Boundary cases checked.

- If $\mathcal{R} = \mathcal{U} = \mathcal{V} = 0$, the equation is linear; the theorem reduces to a trivial terminal-value linear ODE statement.
- If $n = m = 1$, the criteria reduce to scalar inequalities and remain coherent.
- If coefficients are continuous, the proof really gives C^1 , not just absolute continuity.

Ideas tried

1. **Reduction to a standard symmetric Riccati equation.** Idea: mimic the closest paper’s strategy and try to transform the asymmetric equation to a classical Riccati ODE. Obstruction: that reduction in the literature depends on additional structural identities forcing symmetry/reducibility of the nonlinear term; with only “affine in Σ ” available, the equation remains rational and genuinely asymmetric.
2. **Hamiltonian linearization of a nonsymmetric Riccati equation.** Idea: write $\Sigma = YX^{-1}$ and linearize the problem as in standard nonsymmetric Riccati theory. Obstruction: the presence of $R_e(t, \Sigma)^{-1}$, with R_e itself depending on Σ , destroys the standard quadratic form $\Sigma B \Sigma$ needed for linearization.
3. **A priori Lyapunov estimate for $\|\Sigma\|$ on arbitrary horizons.** Idea: differentiate $\|\Sigma_t\|^2$ and look for a dissipative estimate. Obstruction: in full generality, the rational term yields only a quadratic-over-affine bound, and taking absolute values loses the sign structure. This led instead to the smaller but rigorous invariant-tube criterion and, under an extra linear-growth bound, the continuation theorem.
4. **Import indefinite-LQ generalized Riccati machinery.** Idea: use the generalized Riccati/LMI theory from indefinite stochastic LQ control. Obstruction: the present equation is asymmetric and rational, not the standard symmetric generalized differential Riccati equation with pseudoinverse constraints; the imported theorems do not match hypotheses exactly. I therefore used them only as methodological background, not as proof steps.

Invoked results

No external theorem is used as a black box in the proof. The contraction principle and the matrix-inverse estimates needed for the argument are proved directly above, so there is no citation-hypothesis audit to report here.

Self-confidence

3 — I checked the fixed-point and continuation arguments carefully and could not break them, but the result is only a sufficient criterion and there is always some risk of a bookkeeping slip in the constants.

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